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Chapter 1: The Clock Room—Thought Experiment

Abstract

Imagine a sealed room: inside there are many clocks. The person in the room can fully read every clock face but has no channel independent of the clocks to contact the outside world. Someone outside the door asks, "What time is it now?"

He possesses all the readings, yet he does not know the mapping relationship between any given reading and the true time—that is, he does not know whether a clock before him is accurate or fast, nor by how much. This is the epistemological predicament depicted by the "Clock Room" thought experiment.

The core thesis of this thought experiment is: whether a cognizer can infer the true state from readings depends on reliable constraints on the mapping function (π), and cannot be guaranteed by the number of readings alone. When the mapping is completely unknown, the candidate mappings are subject to no prior constraints, and independent calibration is absent, no amount of readings can rule out any candidate true state. Using readings to infer the mapping, then using that inferred mapping to explain the readings, can form an internally self-consistent relational network, yet this self-consistency alone cannot confirm alignment with external reality; even when the mapping is known, unique recovery further depends on whether the mapping itself preserves the requisite differences among states.

To precisely characterize this structural misalignment, this book introduces the conceptual tool of the fiber: the set of all possible true states compatible with a given reading. Within this framework, a non-singleton fiber—that

is, a fiber in which one reading is compatible with more than one true state—is a geometric formulation of the Clock Room dilemma. The fiber framework further identifies three layers of progressive irreversibility: when the mapping is completely unknown and candidate mappings face no prior constraints, the set of compatible states is the full set; prior constraints, where present, may shrink it (information deficit); when readings in turn rewrite the mapping itself, existing calibration constraints face continuous pressure to fail (dynamic corrosion); and, under the usual Euclidean or finite-dimensional manifold conditions, a continuous mapping from a higher-dimensional true state space to a lower-dimensional observation space cannot remain injective across all true states (ontological dimension reduction).

John von Neumann himself stated that his views regarding absolute mathematical truth "changed three times in succession"; this intellectual trajectory offers a historical structural analogy for this predicament: this most precise builder of formal systems in the 20th century once supported the program of proving the consistency of classical mathematics by finitistic methods, and later re-evaluated the grounds of absolute mathematical truth. This trajectory concerns the limits of the methods of proof and is not to be equated with the underdeterminability of states in the Clock Room. The act of mapping not only generates readings but also reshapes the mode of existence of the cognizer—when a person entrusts the rhythm of his actions to a clock, he slips from "existing in time" to "reacting within the clock." This same structure reappears in the contemporary artificial intelligence (AI) alignment problem, constituting a modern version of the "many-readings, unknown-mapping" dilemma.

Reading, mapping, and reality are three distinct epistemological categories: a reading is not reality itself, more readings do not amount to more knowledge; furthermore, the act of mapping itself changes the relationship between the cognizer and reality.

Keywords

The Clock Room; mapping function; fiber; three-layer irreversibility; calibration; John von Neumann

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Pre-reading Note: On the term "fiber"

Section §3 of this chapter will introduce a mathematical concept—"fiber." This chapter takes only its intuitive core:

Fiber = the set of all possible true states compatible with a single reading.

For example: A circular shadow appears on a wall. Behind the wall there might be a sphere, a disk, or a cylinder. This shadow (the reading) does not tell the observer which object is actually behind the wall. The set of all objects that could cast this shadow is the "fiber" of that shadow.

This example will be fully developed in §3.1. This chapter uses the term "fiber" because it can precisely express a fact long perceived but difficult to formalize: **Where observations (readings) do not suffice to determine the true state, behind a single reading there is more than one possibility.** Readers need not know topology or fiber bundle theory; they only need to carry this shadow metaphor while reading on.

§1 The Clock Room

§1.1 The Minimal Setup

The simplest configuration of this epistemological experiment is as follows: A sealed room, without windows. Inside the room are many clocks, each displaying a time reading. The person in the room knows perfectly how to read a dial. He can report the reading of every clock, calculate the difference between any two clocks, and sort them by the magnitude of their readings. Someone outside the door asks: "What time is it now?"

Complete competence and total impotence coexist within the same subject—this is precisely the structural feature of the mapping predicament (here "mapping" refers to the mapping function that transforms the true state into a reading; for details see §3.2). In terms of computational capacity, he can execute any reasoning on the readings; in terms of calibration capacity, he requires independent verification of whether any reading corresponds to the true time—he possesses the former in full and lacks the latter entirely.

Before entering the structure of the Clock Room, a prior question must be answered: Why clocks? Lewis Mumford, in *Technics and Civilization* (1934), gave a precise verdict:

> "The clock, not the steam-engine, is the key-machine of the modern industrial age. For every phase of its development the clock is both the outstanding fact and the typical symbol of the machine: even today no other machine is so ubiquitous." (Mumford, 1934)

More than four decades later, Joseph Weizenbaum, in *Computer Power and Human Reason* (1976), further anchored the ontological status of the clock—clocks were the first autonomous machines built by humans, machines that do not extend human muscle or sense organs but run by themselves on the basis of an internalized model of some aspect of the real world:

> "An autonomous machine is one that, once started, runs by itself on the basis of an internalized model of some aspect of the real world. Clocks are fundamentally models of the planetary system. They are the first autonomous machines built by man, and until the advent of the computer they remained the only truly important ones." (Weizenbaum, 1976)

This is precisely the structural reason for choosing the clock to demonstrate the mapping predicament: the clock as an object is itself the very first "autonomous model that internalizes a true state into a readable number." Using the relationship between the clock and time to discuss "the gap between a reading and reality" is not seeking a convenient metaphor—it is tracing the mapping predicament back to its oldest mechanical carrier.

The following four types of problems show why this predicament is so intractable.

§1.2 How Four Problems Are Generated from the Same Room

The same room generates four distinct types of problems.

Type 1: The Factual Question. Someone outside asks: "What time is it now?"

This is the most direct question. It presupposes that a single unique true value exists, and the person outside expects a definite answer. The person inside faces many candidate readings but has no basis on which to judge which reading corresponds to the true time outside. His predicament is not "no answer," but "no basis for choosing an answer."

Type 2: The Relational Question. Someone outside asks: "How long until six o'clock?"

This question does not demand absolute time, but rather the computation of a time difference: $f(T) = 6:00 - T$. The person inside can, for any clock, calculate how long until six o'clock, meaning he holds as many correct computational results as there are clocks. Yet he still cannot answer—because he cannot choose which clock to use as the starting point. His reasoning capacity is complete; the starting condition is absent.

A variant shifts the question from the absolute moment to duration. If the person outside says: "You can leave in three hours"—this is not a question, but a promise. The promise stipulates the condition of waiting three hours from this moment, yet does not tell the observer whether three hours on the clock face corresponds to three hours of true time. He can record the reading at the moment of the announcement, but cannot confirm when the waiting condition stipulated in the promise has been met. The difficulty is no longer choosing which clock to report "what time it is," but calibrating the duration of the wait. This variant has already slid from the relational question toward the action problem—it is thus the transition to the third type of problem.

Type 3: The Action Problem. Someone outside says: "You can leave at any time. Your reward depends on how close the true time of your exit is to six o'clock sharp."

Now things change. The person is no longer merely asked to produce an answer but must make a decision with consequences. What he needs is not the capacity to read, but judgment for action. And he has no means at all to verify which clock is correct.

Type 4: The Meta-information Problem. While he is still waiting, the door opens, someone pushes in five new clocks, places them on the floor, and leaves. No words are spoken.

The first three types of problems involve the person outside posing an explicit question or laying down conditions for action. In this fourth type, the external agent asks no question; only an event occurs. The person inside must now decide for himself: is this more data, or is it a signal about the source of the data? Does it mean the old clocks are untrustworthy? Are the new clocks more trustworthy? Or is this merely a disturbance?

What distinguishes "treating the event as more data" from "asking what the event itself means" is not intelligence or computational power, but **the capacity to recognize the event itself as a signal needing interpretation—this recognition belongs to metacognition and occurs prior to the processing of the event**. Regardless of the processing outcome[1], signal recognition must happen first. A system that processes information only when questioned exhibits no signal-recognition behavior whatsoever in the face of a Type 4 problem.

These four types of problems illustrate that "the Clock Room" is not a single-point puzzle but a master experiment. The same setup, under different modes of questioning, generates cognitive predicaments at different levels. It is therefore necessary to append a premise: this experiment does not assume whether the person outside the door knows the true time—the person outside might also be situated within a larger sealed space. The argument relies only on the relational structure of information asymmetry between inside and outside the room, and not on either party's possession of the true time.

A person with a complete capacity to read clocks, inside this room, becomes a prisoner incapable of making a well-grounded judgment. He possesses complete computational ability but has lost the grounds for judgment. His predicament lies not in his intellect, but in the structure: the reference required for

[1]The processing outcome can take one of three forms: interrogating the meaning of the event, dismissing the event as noise, or marking it as pending and deferring judgment.

calibration lies outside the system, whereas all the readings from which he might judge the true time come from inside the system, none of them independently calibrated—for the argument see §3.5.

§1.3 Why Intuitive Strategies Fail

Return to the first type of question. Faced with "What time is it now?," the person inside might try the following strategies.

- **Strategy 1: Take a statistic.** Calculate the mean or median of all readings. But to treat these statistics as reliable estimates of the true value requires grounded constraints on the central location and correlation structure of the errors; the mean and the median require different specific conditions. He has no basis whatever for such constraints—a thousand clocks could all be two hours fast, and the mean would still be two hours fast. A statistic does not add information; it only recombines already existing information. If the original information does not contain the true value, the summary still does not.
- **Strategy 2: Look for clustering.** If six hundred clocks are clustered around 3:00, he is inclined to trust this cluster. But clustering reflects consensus, not truth. Six hundred clocks could have come from the same assembly line, sharing the same systematic bias—for instance, all were set two hours fast at the factory. An increase in consensus does not equal the completion of calibration. In extreme cases, consensus is an amplifier of bias, not a signal or indicator of truth. (At the intuitive level, a person facing a cluster cannot distinguish between two possible generative mechanisms—"pointing to the same true value" versus "coming from the same assembly line"—this is precisely the core of the predicament. For the formal argument see §3.6.)
- **Strategy 3: Observe running rate.** Which clocks are running at a constant rate? Which have stopped? Clocks with a normal rate "seem" more reliable. But a correct rate (ticking one second per second) only guarantees the increment is accurate, not that the starting point is accurate. A clock that is two hours fast but has a perfect rate will forever remain two hours fast. Running correctly is not equivalent to pointing correctly.
- **Strategy 4: Select the authoritative clock.** One clock looks more precise than the others, more expensive, labeled "standard time." He is inclined to trust it. But "looking authoritative" is a narrative attribute, not a calibration attribute. Labels can be faked; precision need not amount to accuracy. A clock does not point to the correct time simply because it is gilded.
- **Strategy 5: Admit ignorance.** He says: "I can report the readings of some clocks, but I do not know what time it is now."

The common structure of the first four strategies is: **using a clock to calibrate a clock.** In the absence of an external anchor, using a clock to calibrate a clock is merely a closed loop. It can produce internal consistency, but it cannot guarantee alignment with reality.

The only honest answer is the fifth. "Using a clock to calibrate a clock is not calibration" is the fundamental judgment running throughout this chapter—hereafter this claim will not be re-argued, only cited. (For the question of whether "admitting ignorance" itself is also just an internal reading, see the self-reference warning in §7.1.)

Epistemology in English contains a famous clock-adjacent case—Russell's stopped clock: a man looks at a clock that has stopped, believes it is going, and happens to look at it at a moment when it is right, thereby

acquiring a true belief as to the time of day; Russell uses this to show that a true belief is not yet knowledge (Russell, 1948). The Clock Room differs from the stopped clock in structure. The man in the stopped-clock case holds a belief about the mapping that happens to be false—he believes the clock is going—and his true belief is acquired by luck; the honest person in the Clock Room does not pretend to know the mapping—he knows that no reading has been independently calibrated. The former asks whether a luckily true belief counts as knowledge; the latter asks whether readings can determine the true value when the mapping is unknown—this chapter is concerned with the latter. The literature on justified true belief sparked by Gettier's cases (Gettier, 1963) continues the former question: the crux of those counterexamples is the luck of validly deriving a true conclusion from a false premise, not the unknown mapping between readings and reality.

§1.4 The Piling Up of Clock Quantity (N)

A seemingly natural intuition is: more clocks should provide more information and thus be more helpful. Under the setup of this experiment, however, this intuition must be re-examined. The epistemological difference between one uncalibrated clock and a thousand uncalibrated clocks is not as significant as one might imagine. If they share the same systematic bias, an increase in number only increases redundancy, not independent information about the true time.

If the number of clocks (N) becomes very large—covering every possible reading on the dial—the situation becomes worse, not better. What he faces is no longer "certain candidate answers," but "all possible answers are simultaneously present." For any time he might wish to answer, there is necessarily a clock on the wall pointing exactly to it—but pointing to it does not prove it. The fiber has not shrunk—for the formal argument see §3.5.

If the number of clocks (N) increases to a hundred billion, there is yet another dimension of deterioration. Reviewing measurements spanning the better part of a century and covering perception, action, and imagination, Zheng and Meister state:

> The information throughput of a human being is about 10 bits/s. In comparison, our sensory systems gather data at $\sim 10^9$ bits/s. (Zheng & Meister, 2025)

The paper itself notes that the stark contrast between these two numbers remains unexplained. When the clocks become too numerous to read, the person is forced to construct a "meta-system" for filtering, summarizing, and weighting these clocks. And what is this meta-system itself? **A clock about clocks—a meta-clock.** The construction process of the meta-clock embeds prior assumptions. Some prior assumptions are of higher quality than others, but in the absence of external calibration, the quality of the assumptions themselves cannot be verified. The builder of the meta-clock must choose which assumptions to use without knowing which are correct—the mapping function of this choice is itself unknown. Once the meta-clock is built, it **replaces the original clocks as his operational interface**—he no longer faces the readings directly, but faces the output of the meta-clock. The problem is elevated to another layer of abstraction, but it does not disappear.

The key variable is not the number of clocks (N), but the degree to which the mapping function (π) is known. No matter how large the number of clocks (N) becomes, under the boundary conditions of complete unknownness and lack of external calibration, the mere increase in the number of clocks (N) provides no verifiable gain regarding the true value. This judgment has an independent metric formulation in cybernetics: the regulator's capacity to regulate cannot exceed its own capacity as a communication channel—variety exceeding the channel

capacity cannot be converted into regulatory capacity (Ashby, 1956).

When all clocks display exactly the same reading, the person inside is most likely to simply report that number, without further hesitation. Yet his epistemological situation has not improved—completely identical readings could arise from all clocks sharing the same systematic bias. The mapping function remains unknown, and the fiber has not shrunk; what has changed is only a marked increase in subjective certainty. In this situation, **the observer's motivation to initiate further calibration or doubt is reduced to a minimum—even though the need for calibration has not diminished.**

This is the Clock Room. A minimal setup, four types of problems, four failed strategies, and a predicament that cannot be rescued by the number of clocks (N).

To understand why this predicament is so intractable—that is, the "structural distance between reading and reality" identified in §1—it is necessary first to look at a person, one of the most precise clock-makers of the 20th century, and how he encountered it in his own research and reflection.

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§2 The Most Precise Clock-Maker

The role of this section is not to produce a history of von Neumann's thought, but to use the experience of a worker inside mathematics to illustrate how the goal of seeking the grounds of absolute reliability within mathematical methods met its limits, and to make a bounded structural comparison with the Clock Room.

§2.1 von Neumann's Three Self-Negations

John von Neumann was one of the greatest mathematicians of the 20th century. In 1925, he provided an axiomatic system for set theory. He also laid the rigorous mathematical foundations for quantum mechanics, founded game theory, participated in the development of the first general-purpose electronic computer—the Electronic Numerical Integrator and Computer (ENIAC)—and proposed the design of the modern stored-program general-purpose computer.[2]

He appears in this chapter not because of his fame, but because of his experience. He was one of the "most precise clock-makers" of the 20th century, who, in the course of making "clocks," discovered their limitations. He was not a bystander merely repeating a conclusion reached elsewhere—Gödel's announcement of the first incompleteness theorem and the ensuing discussion provided a crucial intellectual stimulus; von Neumann then used Gödel's methods to derive the second incompleteness theorem independently and reassessed Hilbert's program in its light.

In 1947, he wrote in his essay "The Mathematician":

> "This happened in our own lifetime, and I know myself how humiliatingly easily my own views regarding the absolute mathematical truth changed during this episode, and how they changed three times in succession!" (von Neumann, 1947, "The Mathematician")

[2]For an overview of von Neumann's scientific career, see the translator's preface by Li Wenlin to *The Role of Mathematics in the Sciences and in Society* (2009). The following division of the evolution of his thought into three stages is a heuristic reconstruction. He himself stated in "The Mathematician" (1947) only that his views "changed three times in succession," without giving any periodization; this reconstruction retains a directional correspondence without claiming a strict historical periodization.

The following characterization of the three shifts is based on von Neumann's own writings and correspondence from the relevant periods: the 1925 paper on the axiomatization of set theory; the correspondence with Gödel and Carnap from November 1930 to June 1931; "The Mathematician" (1947); "The Role of Mathematics in the Sciences and in Society" (1954); and "Can We Survive Technology?" (1955).[3]

The First Shift (1925–autumn 1930). Von Neumann stood on Hilbert's side, believing that mathematics could obtain a secure foundation through the axiomatic method. In 1925 he published a paper on the axiomatization of set theory (von Neumann, 1925/1967), attempting to provide a logically unobjectionable formal system for all of mathematics. He believed the aim of Hilbert's program—that the consistency of classical mathematics could be proved by finitistic methods—was achievable. This distinguishes the finitistic mathematics serving as the method of proof from the classical mathematical system under study; the program is not to be identified directly with the studied system proving its own consistency.

The Second Shift (1930–1931). At the 1930 Königsberg conference, Gödel announced the first incompleteness theorem. Among those present, almost only von Neumann grasped the significance of the announcement. A few weeks later, he independently derived the proof of the second incompleteness theorem—that no sufficiently strong, consistent formal system can prove its own consistency. He informed Gödel of this in a letter dated November 20, 1930, but upon learning that Gödel had already submitted his paper, he discreetly abandoned his claim to priority (Formica, 2011; Rédei, 2005; von Plato, 2020).[4]

His assessment of this result was as follows:

> "If a system of mathematics does not lead into contradiction, then this fact cannot be demonstrated with the procedures of that system. … My personal opinion, which is shared by many others, is, that Gödel has shown that Hilbert's program is essentially hopeless." (von Neumann, 1947, "The Mathematician")

Gödel's theorem shares the same structural intuition as the judgment in §1.3. It must be emphasized that the two are not the same mathematical proposition—Gödel's theorem concerns provability, and the proposition of §3 of this chapter concerns state determinability—but under this book's interpretation they share a structural feature: a system cannot, using only its internal resources, ascertain its own relationship to what it attempts to represent. This chapter discusses Gödel's theorem as a conclusion within the same domain as the judgment of §1.3, without claiming the two are equivalent.

The Third Shift (approximately after 1931). Neither Hilbert's program nor the Brouwer-Weyl program could

[3]For the Chinese translation used here, see Li Wenlin, ed., Cheng Zhao, Wang Lixia, and Yang Jing, trans., *The Role of Mathematics in the Sciences and in Society* (Dalian University of Technology Press, 2009). For the correspondence, see Rédei, ed., *John von Neumann: Selected Letters* (American Mathematical Society, 2005). The English originals of "The Mathematician" (1947), "The Role of Mathematics in the Sciences and in Society" (1954), and "Can We Survive Technology?" (1955) are collected in Part VII of *The Neumann Compendium* (Bródy & Vámos eds., World Scientific, 1995). In the Chinese version of this chapter, the first sentence of the quotation on system consistency in §2.1 and the first sentence of the quotation on the scope of theoretical physicists' knowledge in §2.2 are translated directly from the English originals; they are not verbatim quotations from the Chinese translation listed above.

[4]The three key pieces of correspondence are in Rédei (2005); the originals of the first two letters (to Gödel) are in Princeton University's Gödel Archive, the third (to Carnap) original is in the Archives of Scientific Philosophy at the University of Pittsburgh (RC 029-08-01). (1) Von Neumann to Gödel, November 20, 1930 (Rédei, 2005): "I was able to show that the consistency of mathematics is unprovable." Von Neumann never published this result, and his complete proof has not survived (it is absent from both his and Gödel's unpublished materials); what is known today is the sketch of the argument given in the letter, and the literature offers different reconstructions of his actual derivation (Formica, 2023). (2) Von Neumann to Gödel, November 29, 1930 (Rédei, 2005): "As you have established the theorem on the unprovability of consistency as a natural continuation and deepening of your earlier results, I clearly won't publish on this subject." (3) Von Neumann to Carnap, June 7, 1931 (Rédei, 2005): "Thus I am today of the opinion that (1) Gödel has shown the unrealizability of Hilbert's program."

provide an absolute justification for classical mathematics. Most mathematicians decided to use the system anyway. Von Neumann wrote:

> "After all, classical mathematics was producing results which were both elegant and useful, and, even though one could never again be absolutely certain of its reliability, it stood on at least as sound a foundation as, for example, the existence of the electron." (von Neumann, 1947, "The Mathematician")

In his 1947 essay "The Mathematician," von Neumann wrote that his views regarding absolute mathematical truth had "changed three times in succession"; in 1954, in "The Role of Mathematics in the Sciences and in Society," he said that within some thirty years his conviction as to what mathematical rigor is had "changed at least twice." The two self-assessments were not aimed at the same object—the former concerns views on absolute mathematical truth, the latter the conviction as to what mathematical rigor is—and this book draws no conclusion from the difference in counts about whether the "third" shift was completed. In that 1954 essay, he also wrote: "So there is a certain area of doubt there; and in evaluating the character and the role of mathematics one must not forget that doubt exists."

What follows is this book's reading of this turn, not von Neumann's own statement. What he did here is precisely the opposite of Hilbert: Hilbert attempted to internalize the epistemological problem—to prove the reliability of classical mathematics through finitistic mathematics; von Neumann externalized it. Here "internalize" means seeking the grounds of reliability within mathematical methods; it does not amount to proving, within one and the same formal system, that very system's own consistency. The self-assessment above is precisely where the turn lands—classical mathematics "stood on at least as sound a foundation as, for example, the existence of the electron" (von Neumann, 1947)—the justification has shifted from a consistency proof within mathematics to the effectiveness of empirical science. He no longer rested absolute reliability on this proof program, but accepted an ineliminable uncertainty, and then did his work within that uncertainty—while always remembering that the uncertainty existed.

§2.2 The Baroque Diagnosis

Von Neumann not only experienced this difficulty at the level of formal systems; he also diagnosed the same structure at the level of disciplinary evolution.

In "The Mathematician," he observed: mathematics is born from experience, then develops its own formal system. After operating within a formal system for some time, a discipline shows a tendency to degenerate:

> "As a mathematical discipline travels far from its empirical source … [i]t becomes more and more purely aestheticizing, more and more purely l'art pour l'art." (von Neumann, 1947, "The Mathematician")

He used a precise word: **baroque**. He invoked the term "baroque" to describe a precision that loses its functionality through self-referencing—just as baroque ornamentation is layered upon itself, moving away from the structure itself.

> "At the inception the style is usually classical; when it shows signs of becoming baroque, then the danger signal is up." (von Neumann, 1947, "The Mathematician")

The characteristic of the baroque style is that ornament refers to ornament, form refers to form, the structure

becomes more and more refined but further and further from its original function. In the language of the Clock Room, this is the disciplinary version of the closed loop from §1.3: theorems prove theorems, techniques serve techniques, concepts are built upon concepts—the empirical problems that gave birth to the discipline have vanished from the input of the mapping.

He gave a prescription:

> "In any event, whenever this stage is reached, the only remedy seems to me to be the rejuvenating return to the source: the reinjection of more or less directly empirical ideas. I am convinced that this was a necessary condition to conserve the freshness and the vitality of the subject and that this will remain equally true in the future." (von Neumann, 1947, "The Mathematician")

This book further interprets the "empirical source" as an embodied, multi-channel mode of contact; this is not von Neumann's verbatim definition of the term. As used in this chapter, it is not a specific calibration object, but the way human beings, as embodied existents, make contact with the world—the hand touching a material, an experiment producing a surprise, or an observation yielding a phenomenon not predicted by theory. To lose touch with the empirical source is therefore not to lose a particular external reference, but to lose this mode of contact itself. (Bodily perception is itself also a mapping, but it provides multi-channel, cross-verifiable modes of contact—a qualitative difference from single-dimension symbolic operation; see §5.2.)

He used the phrase "whenever this stage is reached"—degeneration is not a one-time event, but a recurring tendency. Researchers repeatedly need to return to the discipline's empirical source, and after each return the discipline drifts away again.

He also noticed a related symptom:

> "A good theoretical physicist may today still have a working knowledge of more than half of his subject. I doubt that any mathematician now living has much of a relationship to more than a quarter." (von Neumann, 1947, "The Mathematician")

Why can a physicist master half the knowledge of his field? Because physics, tightly constrained by empirical checks, keeps knowledge in a concentrated state—"at almost all times most of the effort of all theoretical physicists was concentrated on no more than one or two very sharply circumscribed fields." In von Neumann's 1947 comparison, the problems of theoretical physics were usually given objectively by experimental difficulties, whereas the branches of mathematics allowed researchers greater freedom of topic selection; compared with theoretical physics, mathematics generally lacked an equally direct and unified empirical constraint. In the branches remote from their empirical sources that he was discussing, new directions were governed first by internal consistency and aesthetic criteria, but this does not mean that mathematics as a whole has no empirical constraint.

§2.3 What This Means for the Clock Room

Von Neumann's three shifts are not philosophical speculation—the impetus of the shifts includes the formal operations he executed with his own hands and their consequences. They provide a structural analogy, from within research on formal systems, for the judgment of §1.3 (see the boundary statement in §2.1): one of the most precise clock-makers of the 20th century, in the course of sustained mathematical research and its reflection, moved from "believing that the consistency of classical mathematics could be proved by finitistic methods" to

"accepting the existence of uncertainty, and doing work within that uncertainty." It was not because he was not smart enough. It was because the methods available to consistency proofs are themselves limited.

This chapter additionally places two judgments from different contexts in a bounded comparison with the Clock Room. In 1954, in "The Role of Mathematics in the Sciences and in Society," he discussed the long-term effectiveness of research not governed by short-term utilitarian aims:

> "Successes were largely due to forgetting completely about what one ultimately wanted, or whether one wanted anything ultimately; in refusing to investigate things which profit, and in relying solely on guidance by criteria of intellectual elegance; it was by following this rule that one actually got ahead in the long run, much better than any strictly utilitarian course would have permitted." (von Neumann, 1954, "The Role of Mathematics in the Sciences and in Society"; the Chinese edition quotes this sentence from the Chinese translation edited by Li Wenlin, which omits the clause "or whether one wanted anything ultimately"; the English edition quotes the original sentence in full, and the difference in coverage is noted here)

In 1955, in "Can We Survive Technology?," he gave a safety rule:

> "Any attempt to find automatically safe channels for the present explosive variety of progress must lead to frustration. The only safety possible is relative, and it lies in an intelligent exercise of day-to-day judgment." (von Neumann, 1955, "Can We Survive Technology?")

The two passages belong to different contexts. The 1954 passage addresses the time lag between scientific discovery and application and argues that research not governed by short-term utilitarian aims can yield long-term results. The 1955 passage addresses the safety of technological progress and argues that automatically safe channels are unavailable, leaving safety dependent on day-to-day judgment. Connecting them with the Clock Room is this book's further interpretation, not a causal or intellectual-historical sequence established by von Neumann. The comparison proposed here is that research outcomes are not guaranteed by ultimate purposes, and safety is not guaranteed by automatic systems; neither removes the responsibility of judgment in concrete situations. The 1954 phrase "forgetting completely about what one ultimately wanted" cannot be directly equated with abandoning mathematical foundational proof.

To state precisely the predicament to which this prescription responds—to transform the intuitive judgment of §1 into a formalized proposition—requires a language more rigorous than narrative. The next section provides this language with the fiber.

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§3 From Shadow to Fiber

§3.1 The Shadow Metaphor

Imagine a wall. On the wall there is a circular shadow.

Question: What is the shape of the object that casts this shadow?

The observer answers: It could be a sphere. It could be a disk. It could be a cylinder. It could also be some other object that, at a certain angle of illumination, happens to cast a circular shadow.

The shadow is a reading. The object behind the wall is the reality. The core problem of the shadow metaphor is that "the same reading can correspond to many different true states." The direction of the light determines how the object is mapped into a shadow—it is the "mapping function" in this projection scenario.

This metaphor is structurally similar to Plato's Cave, but the focus differs—the cave allegory is concerned with whether the prisoners can turn around and see the truth; this chapter's metaphor is concerned with the fact that even if the observer is fully awake, a single shadow is insufficient to determine the object.

§3.2 Fiber: A First Explanation

Before entering a formal definition, let one term be clarified: **This book uniformly refers to "the correspondence from true states to readings" as the "mapping function" (π).** In the Clock Room, each clock is a mapping function—it converts the true time into a dial reading. The mapping function might be accurate, might be two hours fast, might run sometimes fast and sometimes slow. The key point is: the person inside the room does not know the mapping function.

Given a reading, the set of all true states that could produce that reading is called, in mathematics, the **fiber**.

In this chapter, "fiber" can first be read as:

> the set of all possible true states compatible with a single reading

In the shadow metaphor, it is:

> all objects that could cast this shadow

Within the same task and under the same criterion of comparison, if new evidence shrinks the set of candidate true states by set inclusion, the reading has ruled out more candidate states; for sets not related by inclusion, information cannot be inferred from 'size' alone.[5]

How this framework systematically distinguishes three types of predicaments of different natures will be fully developed in §4.

§3.3 Translation Table

Table 1. Correspondences between Fiber Language and the Shadow Metaphor

Fiber Language	Shadow Metaphor
Fiber	All objects that can cast the same shadow
Fiber is the full set	This shadow could be caused by anything
Fiber is a singleton	Only one kind of object can cast this shadow
Fiber shrinks	Some possible objects have been ruled out
Fiber is non-singleton	One shadow corresponds to multiple possible objects

[5]When using the term "fiber" in this chapter, its core intuition is taken—the set of all possible true states compatible with a single reading. The full apparatus of fiber bundle theory is not needed here, only the intuition of "how many possible true states correspond to one reading." Readers who require mathematical precision may refer to Steenrod (1951). This translation vocabulary itself is also a clock—it is a mapping of the fiber concept, not the fiber concept itself.

§3.4 Formal Definitions

Definition 1 (True State). There exists a true state T, not directly observable by the person in the room.

Definition 2 (Mapping Function). Each clock is a mapping function π_i: $r_i = \pi_i(T)$. The mapping function encodes all the characteristics of the i-th clock. The person in the room does not know the mapping functions. He only sees the readings r_i.

Definition 3 (Fiber). Given a reading r, the fiber is:

$$\pi_i^{-1}(r) = \{T' \mid \pi_i(T') = r\}$$

The formula above concerns a given mapping. When the mapping is unknown, this chapter uses "fiber" to denote the set of true states compatible with the existing readings under the not-yet-excluded candidate mappings; added calibration knowledge shrinks this candidate set. The structural non-singleton character discussed in §4.3 is a property of the given mapping itself.

Definition 4 (Calibration). Calibration is the process of acquiring knowledge about the mapping function (π_i), thereby shrinking the fiber (the range of possible true states). Calibration requires access to an independent reference in causal contact with the true state. Independence here means that the reference evidence is not generated circularly by the set of readings being calibrated and that its errors do not share their source with the system being calibrated; it does not depend on whether the reference is physically outside the room or formally belongs to some larger set of readings.

§3.5 Proposition and Proof

The following proposition concerns the boundary condition in which the mappings are completely unknown and the space of candidate mapping functions is subject to no prior constraints. The effect of prior constraints on shrinking the fiber under realistic conditions is discussed immediately after the proof.

Proposition. If all mapping functions $\{\pi_1, \dots, \pi_N\}$ are unknown and the space of candidate mapping functions is subject to no prior constraints, then for any observation set $\{r_1, \dots, r_N\}$, all possible true states T' are compatible with the observations.[6]

Proof. Take an arbitrary T'. For each i, construct π_i' such that $\pi_i'(T') = r_i$. Since the space of candidate mapping functions is subject to no prior constraints, π_i' is a legitimate candidate. Therefore, T' is compatible with all observations. ■

Here, "no prior constraints" is a boundary condition, not a typical condition. This boundary condition has a name in the philosophy of science: underdetermination—evidence (observations) insufficient to determine

[6] A limited structural analogy can be drawn between this section's proposition and Hume's "problem of induction" (*A Treatise of Human Nature*, Book I, Part III, "Of Knowledge and Probability", Hume, 1739–1740): existing observations do not by themselves guarantee all the premises needed to move from observation to conclusion. Hume asks why observed causal connections may be extended to unobserved cases; this section asks why readings cannot determine the true state when the candidate mappings are subject to no prior constraints. The objects of the two questions and the conditions under which they can be remedied differ; this chapter does not treat adding an external calibration channel as a solution to Hume's problem. This chapter does not develop this analogy: Hume's discussion is set in the field of causal laws ("future events resemble past events"), whereas this section is set in the field of "mapping functions between readings and reality"; the philosophical loads carried by the two languages are different. The substantive invocation of Hume in this book is concentrated on his second problem (the is–ought problem), reserved for the epilogue's discussion of "the existential conditions of the human as human"; here only this limited structural analogy is acknowledged and ceded.

theory uniquely. The proposition of this section is the form this structure takes between readings and true states. In reality, people always know something about mapping functions: it is a clock, not a thermometer; it is roughly monotonic; its running rate has physical limits. These prior constraints can shrink the fiber, sometimes to a degree perfectly sufficient in practice—people consult their watches to decide when to set out each day.

The thesis of this chapter is not that "priors are useless," but two points: First, priors can shrink the range of candidate true states—under specific assumptions, down to a single candidate. But this uniqueness is conditional on the prior holding; when the prior itself has not been independently calibrated, uniqueness of the candidate state does not by itself confirm that state as the true state. Second, and more importantly, the external reliability of a prior is not guaranteed by internal consistency alone. Some priors can be ruled out by internal observation; but not being ruled out does not mean being calibrated. The assumption "most clocks are accurate," if it happens to be correct, will cause the median to approximate the true value—but even when several clocks give consistent readings, that consistency cannot by itself rule out a common bias—this is the crux of the closed-loop pseudo-calibration in §3.6. The effectiveness of a prior in shrinking the fiber depends on the accuracy of the prior itself, and for a prior not ruled out internally, its external accuracy remains unknown absent independent calibration. A prior that happens to be correct shrinks the fiber to an operable range; a systematically wrong prior shrinks the fiber to a wrong interval far from the true value—the latter is more dangerous than the fiber being the full set, because it manufactures false certainty.

The core conclusion remains unchanged: The key variable is not the quantity of data, but the degree to which the mapping function is known. The role of this proposition is not to describe how people always cognize the world in daily life, but to reveal: once any reliable grasp of the mapping function is lost, why merely proliferating readings cannot substitute for calibration. (§4.1 discusses an institutional case that entered legal proceedings: verification was carried out to a degree unprecedented at the time, and one must still distinguish whether the proof was completed, what kinds of argument are accepted as proof, and whether the model was externally calibrated. Progress in in-chain verification cannot substitute for the last of these.)

§3.6 Closed-Loop Pseudo-Calibration

The person in the room can calculate the difference between any two readings. But this yields only relative relationships among readings, not the absolute relationship between any reading and the true state. He can construct a complete table of differences—an internally consistent relational structure—but it is a coordinate grid without an origin—a coordinate system with only relative relations and no absolute benchmark. A system in which all clocks are two hours fast but have constant mutual gaps is internally perfectly consistent, and externally entirely wrong.

An external observer (such as the person outside the door), seeing only the internal consistency, often misjudges the system as calibrated. But internal consistency is insufficient to prove alignment with reality. The person inside the room may also fall into this misjudgment; since he cannot access an independent calibration signal, this misjudgment lacks the basis for being corrected. A fully enclosed echo chamber can have perfectly harmonious internal sound, while external reality is completely absent.

§3.7 Connection to Measurement Theory

Representational measurement theory (Krantz et al., 1971) defines measurement as a homomorphic mapping from an empirical relational system to a numerical relational system. The model-based view of measurement (Tal, 2020) further points out that readings are not directly "read off" from the object, but are generated in a complex mediated process involving models of the instrument, the environment, and calibration.

The incremental contribution of this chapter is to extend this insight from physical measurement to practical situations that rely on mapping functions to generate readings—AI alignment, financial indicators, and academic evaluation; to provide a unified language through the intuitive framework of the fiber; and to organize unknown mappings, dynamic corrosion, and dimension-reduction limits into a three-layer progressive structure. Classical measurement theory already analyzes many-to-one homomorphisms; this chapter therefore does not claim that the non-singleton character of layer 3 is itself unprecedented in classical measurement theory. Its increment lies in integrating the three problems and in their diagnostic use in settings such as AI alignment.

There is a precedent for using fiber bundle constructions in the field-theoretic analysis of the measurement process (Urbański & Samsonowicz, 2003), but their philosophical orientations differ: in that paper the base space consists of topological properties of the measured quantity and the fiber points toward algebraic properties of the scale values, modeling the topological structure of the physical interaction between the measuring instrument and the measured object; in this chapter, the base space is the reading space and the fiber points toward the set of possible true states compatible with the reading, modeling the epistemological structure of mapping irreversibility between the cognizer and reality. The two share a mathematical metaphor but study different objects. Tal's review already discusses economic measurement and the limitations of single metrics for multi-faceted social concepts; this chapter's contribution is not to introduce measurement analysis into these fields for the first time, but to integrate these problems with unknown mappings, dynamic corrosion, and the limits of dimension reduction within a common diagnostic framework.

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§4 The Three Layers of Irreversibility

§4.1 Layer 1: Information Deficit

Under the condition that the mapping function is unknown, the fiber (the set of possible true states compatible with a given reading) is too large; the information provided by the reading itself is insufficient to uniquely determine the true state.

This layer is solvable in principle. If one can establish an information channel that is independent of all clocks and has causal contact with the true state, one can obtain the first calibration point. The first calibration must come from an evidential chain independent of the set of readings being calibrated—the calibration chain must have one end anchored outside the circular generation of that set.

"Solvable" here does not mean attaining absolute reality. The external reference itself is also a mapping; calibrating through it cannot guarantee unique recovery of the true state. "Solvable" means the fiber shrinks from the full set (or a very large set) to a degree sufficient to support action decisions. This shrinkage constitutes a critical change at the practical level. The solvability of layer 1 depends on the independence of the calibration channel—it must be in causal contact with the true state (Definition 4), but its evidence must not be generated

circularly by the set of readings being calibrated, nor share its source of error with that system. When this independence is eroded, the solution for layer 1 degrades into the persistent competition of layer 2.

One corollary must be made explicit here: appending internal verification does not change the situation of layer 1. §1.4 showed that the quantity of readings does not produce an anchor; what must be added here is that the rigor of verification likewise does not produce an anchor. This claim requires a qualification to be accurate: in-chain verification does shrink the fiber—it monotonically shrinks the fiber "between design and specification," that is, whether the design conforms to the written specification; but it does not by itself guarantee a reduction in the fiber "between device and intent," that is, whether the object built conforms to what was originally wanted. The two are different sets, and the shrinkage of the former does not entail the shrinkage of the latter. So the proposition of this section is not that rigorous verification is useless, but that rigorous verification acts on a different fiber. Increasing the quantity and precision of mutual comparisons within the set of readings can enhance the internal consistency of the set, but cannot generate an anchor point within the set. The qualification for a calibration point is determined not by the rigor of verification, but by whether the verification evidence is independent of the circular generation and shared error sources of the set being calibrated.

The circularity of standards. Suppose one uses a mercury thermometer to measure the temperature of an object, with the aim of testing the law that "objects expand when heated." Here there appears to be a vicious circle: the mercury column indicates temperature precisely by expanding when heated; using it to test the law of thermal expansion amounts to using the law to be tested to test the law to be tested (Franklin et al., 1989, cited in Tal, 2020). This circularity is not unique to thermometers. Using theory to determine which measurement standard is more accurate is equally circular—because before a standard is selected, theory cannot be definitively applied to individual objects. This circularity already has names in the philosophy of science: van Fraassen calls it the problem of coordination, and Chang calls it the problem of nomic measurement (the problem of coordination comes from van Fraassen 2008, the problem of nomic measurement from Chang 2004; both cited in Tal, 2020's review). What this section calls layer 1 is the form this problem takes in the language of the fiber.

The qualification for a solution comes from the channel, not from precision. Franklin et al.'s conclusion is that this circle is not vicious: a mercury thermometer can be calibrated by another thermometer—one whose working principle does not presuppose the law of thermal expansion, such as a constant-volume gas thermometer—thereby establishing the reliability of the mercury thermometer on an independent basis (cited in Tal, 2020). The key is not that the second thermometer is more precise, but that its working principle does not presuppose the law under test. In the language of this section: the source of its readings does not lie within the set of readings being calibrated, so it can serve as the external reference required by Definition 4 in §3.4. Adding more thermometers of the same kind, or engraving the scale more finely, does not produce this qualification; switching to a channel with an independent principle does.

Nor need the dispute over criteria remain permanently suspended. Standards can change through the mutual refinement of theory and instrumentation; this book calls the following change in the definition of the kilogram, from a physical prototype to a fundamental physical constant, a "change of anchor": Chang describes the history of standards as a series of "epistemic iterations," each iteration respecting the existing tradition while revising it, with successor standards deemed more accurate than their predecessors (cited in Tal, 2020). This process has a datable landing point: the kilogram was redefined in 2019—no longer pointing to any particular kind of physical system, but defined by fixing the value of a fundamental physical constant (Tal, 2020). Calling this change in the definition of the kilogram a "change of anchor" is this book's further interpretation; the term does

not mean that a change of definition by itself completes independent calibration, or that it can replace the mutual refinement of theory and instrumentation.

Why the kilogram had to change its anchor, metrologists themselves calculated: the International Prototype of the Kilogram, as a physical artifact, is not linked to any natural invariant, and its mass appears to be changing relative to the cohort of platinum-iridium standards of the same age, by about 50 micrograms per century, possibly significantly more. More importantly: the consistency among the world's one-kilogram platinum-iridium standards has been achieved to within a few micrograms, with mutual comparison precision even finer, yet the drift of this entire cohort of standards relative to some natural invariant remains unknown at the scale of half a century or even a century, below 1 milligram (Mills et al., 2005). Microgram-level internal consistency does not prove that the entire cohort of standards also possesses microgram-level stability relative to a natural invariant—the coordinate grid without an origin in §3.6 here acquires a quantitative form. The criterion shifted from a physical artifact to a natural constant; it was not moved there by appending precision within the chain, but by changing the anchor.

The dispute over criteria went to court. The Royal Signals and Radar Establishment (RSRE), under the UK Ministry of Defence (MOD), developed a Verifiable Integrated Processor for Enhanced Reliability (VIPER), whose proof chain consisted of mathematical models at four levels of descending abstraction strung together—top-level specification, major state machine, block level, gate level, each level being a model (MacKenzie, 2001); the testing, simulation, inspection, and mathematical analysis it underwent were unprecedented at the time (MacKenzie, 2005). In early 1991, Charter Technologies, a company that had obtained the technology license, sued the Minister of Defence, alleging that the claim "proved" constituted a misrepresentation; the dispute was not about what work had or had not been done, but about whether the product of this process counted as a proof (MacKenzie, 1991, 2001).

As far as actual deployment in MOD safety-critical systems is concerned: according to the Ministry of Defence's written parliamentary answer of May 1991, the processor was at that time "not currently used in any safety-critical computer systems controlled by MOD"; the same answer also acknowledged that there was debate in the theoretical community as to the extent to which the processor could be said to be "proved" in the formal mathematical sense (House of Commons, 1991). The case never went to trial—Charter Technologies went bankrupt before the hearing, and the lawsuit lapsed accordingly (MacKenzie, 2001). Bishop Brock and Warren Hunt, commissioned by the National Aeronautics and Space Administration (NASA) to evaluate this work, also concluded that, as of the report's assessment, the chain's formal verification had not been completed (Brock & Hunt, 1990/1991). John Kershaw of the RSRE team himself thus arrived at an explicit relativist conclusion[7]:

> "Proof," he wrote, "is a typical Humpty Dumpty word, which means precisely what you want it to

[7]Humpty Dumpty is a character in Chapter 6 of Carroll's *Through the Looking-Glass*. Original text: "When *I* use a word," Humpty Dumpty said in rather a scornful tone, "it means just what I choose it to mean—neither more nor less." The ensuing exchange pushes the point further: "The question is," said Alice, "whether you *can* make words mean so many different things." "The question is," said Humpty Dumpty, "which is to be master—that's all." (Carroll, 1871) The usage named after him refers to a word's meaning being stipulated on the spot by the user, with no external standard available for adjudication. The closing of Kershaw's remark, "no more and no less," is an echo of the first sentence; and his point in invoking this allusion lies in the second sentence—the dispute is not over whether a word can be polysemous, but over who has the authority to rule on word meaning, and the criterion thereby remains with the acceptance of practitioners. The same judgment appears earlier and more systematically in the Chinese classical tradition: *Xunzi* "Rectifying Names" (Watson trans.): "Names have no intrinsic appropriateness. One agrees to use a certain name and issues an order to that effect, and if the agreement is abided by and becomes a matter of custom, then the name may be said to be appropriate, but if people do not abide by the agreement, then the name ceases to be appropriate." The difference is that Humpty Dumpty stipulates on the spot as an individual, while Xunzi speaks of collective convention—Kershaw's "most practitioners in the field accept" falls on the latter, not the former.

> mean. … In practice proof seems to mean 'an argument that most practitioners in the field accept as valid,' no more and no less." (Kershaw, quoted in MacKenzie, 2001)

The following inference is made by this book; MacKenzie does not state it thus; the sources cited within this paragraph refer only to the origin of materials. In the language of this chapter: each level of this processor's proof chain lies within the set of readings; model-to-model verification, no matter how much is appended or how rigorously conducted, cannot by itself accomplish an independent calibration between the model and the physical device. Brock and Hunt criticized the two informal translations in the chain, whose correctness could not be proved at the time—what was missing was a formal theory relating the different languages; they also adduced the verification conducted under a uniform logical theory as a contrast, showing that this proof gap concerns whether proof obligations have been discharged (Brock & Hunt, 1990/1991; MacKenzie, 2001). Appending in-chain verification can remedy the proof gap between models, but remedying that gap does not produce a calibration point outside the models.

A distinction must be drawn here between whether a proof has been completed, what kinds of argument are accepted as proof, and whether the model has been externally calibrated. Absent a shared commitment on the criterion of proof, appending evidence within one criterion does not guarantee ending the dispute over the criterion itself; but once the criterion is settled, completing the in-chain proof obligations can still answer whether that criterion is met. Even when that question is answered, the independent calibration between model and physical device, and between specification and intent, is not thereby automatically completed. This processor dispute was never adjudicated by a court on the criterion of proof; it ended with the company's bankruptcy, and the lawsuit gave no answer to the dispute over criteria.

Nor did the criterion thereby disappear; it merely changed carriers: the safety-critical software procurement standard issued by the UK Ministry of Defence in 1991 wrote the criterion into the institutional text—it required formal proofs that could be independently checked by a machine, rather than conventional rigorous arguments (MacKenzie, 2001)[8]. This points to the same kind of need as the bookkeeping structure to be discussed in §7.3 of this chapter: to record separately the criterion of proof adopted, the proof obligations not yet discharged under that criterion, and the external calibration that in-chain proof cannot replace.

Intent and silicon have different limitations in their checking; the criterion remains Definition 4 of §3.4. Silicon satisfies this criterion—not because "the actual behavior of the silicon" is an object that can be seen (what can be obtained is only a finite number of readings produced via test apparatus under finite conditions), but because the source of these readings lies outside the chain, in causal contact with the device, not in deductions between models; in Cohn's terms, a physical device can only be observed and measured, not verified (Cohn, 1988). The architect's intent likewise cannot be directly verified by proofs internal to the specification: in Cohn's terms it is not directly accessible—a specification can express the intent, but whether the specification faithfully and adequately expresses the intent cannot be guaranteed by proofs internal to the specification; the compression from intent to specification is of the nature of layer 3 (§4.3) and does not fall within the range covered by the solution of layer 1.

[8]This standard is the UK Ministry of Defence's *Interim Defence Standard 00-55: The Procurement of Safety Critical Software in Defence Equipment*, Part 1, Issue 1, issued April 5, 1991 (bibliographic information per MacKenzie, 1991, entry 15; this book did not directly consult the original standard). MacKenzie records that what the Ministry of Defence envisioned in this standard was proof produced by an automated theorem prover and then checked by an independently written program; this kind of independent mechanical checking is conceivable only with formal proofs, and almost unimaginable for conventional rigorous arguments (MacKenzie, 2001). The "rigorous argument" as defined by MacKenzie refers to arguments accepted as proofs by mathematicians or relevant experts, but which are not formal proofs (MacKenzie, 2005).

§4.2 Layer 2: Dynamic Corrosion

Assume layer 1 has been partially resolved: a certain clock or indicator is reliable at the initial moment, with its mapping function roughly known and verified.

Under this condition, a new difficulty emerges: once the reading is used for rewards, punishments, resource allocation, or performance evaluation, it may induce actors to adjust their behavior around the reading. Under sufficiently strong and persistent incentives, the mapping function may cease to be stable—its own use participates in rewriting it.

Lucas (1976) gave an influential formalization of this phenomenon in economics: a change in policy induces an adjustment in the decision rules of economic agents—agents re-optimize their strategies in response to the new policy—so model relationships estimated under the old policy regime cannot unconditionally be treated as invariant under policy. Lucas's proposition was originally aimed at macroeconometric models, but its structure has been repeatedly confirmed across a much broader range of domains.[9]

Specifically, there are two mechanisms of drift:

- **Strategic Gaming (Intentional).** Actors actively manipulate the reading to obtain more favorable results on a given indicator, without changing the underlying reality the indicator is supposed to measure. For example, scholars form citation cartels to mutually inflate citation counts; firms smooth quarterly earnings through accounting techniques. In this process, the mapping from the underlying reality to the reading is deliberately modified, while the underlying reality itself remains unchanged.
- **Selective Adaptation (Unintentional).** Actors who do not adapt to the indicator are gradually eliminated from the system during its operation, and survivors exhibit a natural concentration of characteristics adaptive to the indicator. In this case, scholars who do not invest effort in indicator-optimizing behaviors find the visibility of their research outcomes on indicators diminished, placing them at a disadvantage in resource allocation, and the conditions for their continued participation in research gradually vanish. No agent needs to consciously manipulate the indicator for the system to undergo corrosion. Selective adaptation is harder to defend against than strategic gaming—because there is no identifiable acting subject to pinpoint and punish.

Adjacent Support from Five Research Traditions. Goodhart (1984), starting from monetary policy experience, pointed out that a monetary statistic made a policy target loses its properties as a good indicator; Strathern (1997) generalized this judgment into the version widely circulated today.[10] Campbell (1979), starting from social evaluation practice, pointed out that the more a quantitative social indicator is used for social decision-making, the more it is subject to corruption pressures and the more likely it is to distort and corrupt the social processes it is intended to monitor. Espeland and Sauder (2007), starting from empirical research on law school rankings, identified two mechanisms triggering reactivity—"self-fulfilling prophecy" and "commensuration"—and listed the diffusion of gaming strategies as one of the three types of reactivity effects. Gaming strategies

[9] Lucas, R. E., Jr. (1976). Econometric Policy Evaluation: A Critique. *Carnegie-Rochester Conference Series on Public Policy*, 1. The key passage reads: "[G]iven that the structure of an econometric model consists of optimal decision rules of economic agents, and that optimal decision rules vary systematically with changes in the structure of series relevant to the decision maker, it follows that any change in policy will systematically alter the structure of econometric models." The "Lucas critique" as a methodological proposition extends well beyond macroeconometrics itself: this section treats it as one formalized anchor for the destabilization of mapping functions in indicator/reading systems once they are used for policy or evaluation, without making a claim of historical priority.

[10] The widely circulated phrase "When a measure becomes a target, it ceases to be a good measure" is not the original text of Goodhart (1984), but Strathern's (1997) generalized paraphrase of Goodhart's original idea. Goodhart's original formulation was narrower, limited to the domain of monetary policy.

correspond to the strategic gaming (intentional drift) distinguished in this section. A self-fulfilling prophecy confirms itself through expectation-feedback loops, whereas selective adaptation changes population composition through differential survival; the two cannot be equated. Scott (1998), starting from the state's project of legibility, pointed out that "[c]ertain forms of knowledge and control require a narrowing of vision," which renders the central phenomenon more readable; when such simplification is allied with state power, the informal processes thereby suppressed are not merely overlooked but remade. Zhao Dingxin (2022), from the angle of social ontology, pointed out that physical laws do not change their mode of operation because of human understanding, while "the tremendous strategic capacity for action and capacity for self-justification of human beings have led to large-scale disjunction between structure and function in human society, led to unstable positive feedback mechanisms dominating individual human behavior and the development of entire societies, and given human beings a tremendous capacity to create social mechanisms and to change their importance and modes of operation…" (author's translation). These five traditions come from monetary policy, social program evaluation, organizational sociology, state governance, and social-mechanism theory, but they do not each prove the two drift mechanisms distinguished in this section, nor do they constitute five independent pieces of evidence for the same conclusion. Together they support only a broader proposition: when readings, classifications, and evaluations enter processes of action and governance, social actors and their environments change accordingly. Goodhart, Campbell, and Espeland and Sauder more directly support indicator destabilization and reactivity; Scott and Zhao provide adjacent accounts of state-legibility-driven remaking and the instability of social mechanisms, respectively. The selective adaptation defined in this section still requires separate evidence of differential survival and changes in population composition; it cannot be inferred merely from self-fulfilling prophecy or generic positive feedback.

Self-Referencing of the Mapping. When multiple clocks calibrate each other—each clock's reading taking other clocks' readings as its reference—the mapping function degenerates from "a mapping function taking the true state as input $\pi_i(T)$" into "a mapping function taking other readings as input $\pi_i(\{r_j\}_{j\neq i})$." The true state partially vanishes from the input of the mapping. Clocks begin to point more at other clocks than at reality. In this process, the internal consistency of the system continuously grows, while the deviation from reality can also simultaneously grow—the two are not in a trade-off relationship but can advance in parallel. The price discovery mechanism of contemporary financial markets, the citation network of academic evaluation, and social media topic trends all operate, to various degrees, relying on this structure.

Financial markets provide a well-documented instance of this structure. According to the account in Chapter 6 of MacKenzie (2006), after the Black-Scholes-Merton option pricing model entered trading practice, traders switched to using a quantity called implied volatility for communication and pricing—it is obtained by running the model in reverse, using observed option prices to back out the underlying stock's volatility; eventually, option prices were sometimes no longer quoted in dollars but directly in implied volatility (same book, Chapter 9). And this parameter required by the model is not directly observable, so there is no unique theoretical price to compare with the actual price. The missing anchor is ultimately filled by a quantity back-calculated from readings: the number quoted, compared, and used as the basis for decisions is the product of readings processed through the model, not an independent contact with the underlying asset. In the language of this section, the input to the mapping function (π_i) has switched from the true state to other readings; in the language of §1.4, this is a reading-about-readings replacing the original reading as the operational interface.

MacKenzie's own core concept differs in direction from this section. What he calls Barnesian performativity

refers to the actual use of a piece of economics making economic processes more like what economics depicts, making the model more true (same book, Chapter 1)—that is a decrease in deviation, not the increase in deviation discussed in this section. In this book's terms, two situations must be distinguished: the use of a model changing the real process does not automatically invalidate independent checks of the actual outcomes; but treating the agreement produced by that feedback as evidence that the model has been independently calibrated mistakes an outcome partly shaped by the model for independent validation. The reference loses its independence only when the calibration evidence is also generated by this closed loop of readings and independent checks are absent. In that case, the anchor is not drifting; it is being absorbed into the reading system's self-validation. He also points out that this effect does not depend on anyone believing in the model—once the model is incorporated into algorithms, programs, routines, and material devices, it continues to produce effects even if users doubt its merits, do not understand its details, or are even unaware of its existence. This shows that the model's effects need not presuppose the users' belief in the model; to further establish the selective adaptation defined in this section, one must still show how the model participates in the filtering of resource allocation and survival conditions, such that non-adapters gradually exit and the composition of survivors changes accordingly.

What needs to be clear is: self-referencing is a matter of degree, not a categorical issue. Arbitrage mechanisms and bankruptcy events still exist in financial markets—these constitute causal contact with reality. Academic citations still retain some connection to paper content, it is just that this connection may be weakened by strategic citation and reputation-accumulation effects. "Partial decoupling" means calibration is still possible, though more difficult; "full decoupling" means calibration is in principle no longer possible. This chapter discusses the former—the persistent worsening trend of decoupling—rather than the already completed state of the latter.

The Time Cost of Dynamic Reading. Observation itself consumes time. When the person in the room begins to read the first clock, "now" has already become the "past." By the time he finishes reading all the clocks, the true time has advanced further. What he acquires is not "the set of readings at this moment," but a splicing of readings from different moments—a signal combination that lags behind the true state.

Distinction from Layer 1. Layer 1 is a static predicament: the mapping function is unknown but stable, and the solution is a one-time calibration. Layer 2 is a dynamic predicament: the mapping may be rewritten by its use, and once the results of calibration enter processes of incentive and resource allocation, they may trigger a chain reaction of de-calibration. So long as these conditions of incentive and adaptation persist, this is not a problem that can be ended by a one-time intervention, but a process requiring sustained investment of calibration resources.

§4.3 Layer 3: Ontological Dimension Reduction

The first two layers still retain an ideal limit—if the calibration speed were infinitely faster than the corrosion speed, one could perhaps approach perfection. Layer 3 negates this limit.

The true state has multiple dimensions; the instantaneous position in time is only one of its dimensions. The clock compresses a multidimensional reality into a one-dimensional reading, losing the relationships with the discarded dimensions—these relationships are sometimes part of the information of time (e.g., "What time is it?" in a certain context is actually asking "Is it evening or late night?", information that comes from light, bodily state, not from the clock face). Temperature, the angle and color temperature of light, the body's degree of fatigue, sounds outside the window, the humidity of the air, the rhythm of the heartbeat—these are not "ad-

ditional attributes" of time; they are constituent dimensions of "the present moment" itself (as a category of experience; for the physical timekeeping category see below). When the clock compresses "now" into a single number, it performs a dimension reduction. The discarded dimensions can no longer be uniquely recovered from the reading; this does not mean that they have no effect on the reading.

If the dimension of the true state space ($\mathcal{T}$) is higher than the dimension of the observable space ($\mathcal{O}$) (i.e., $\dim(\mathcal{T}) > \dim(\mathcal{O})$), then, under standard topological assumptions (with $\mathcal{T}$ and $\mathcal{O}$ taking their usual forms, such as open regions of Euclidean space or finite-dimensional manifolds), no continuous mapping $\pi : \mathcal{T} \to \mathcal{O}$ can be injective—two distinct true states in the high-dimensional space must map to the same reading in the low-dimensional space. There is therefore at least one non-singleton fiber. Even if one knows every detail of the mapping function perfectly, there exists no inverse mapping capable of uniquely recovering all true states—the information required for globally unique recovery of all true states is no longer reversibly preserved by the mapping, although the reading may still carry partial information about those states.

What the observable space ($\mathcal{O}$) here refers to must be clarified. It refers to low-dimensional continuous objects such as the clock face—the position of the pointer itself varies continuously, so it is meaningful to speak of its dimension and whether the mapping is continuous. The reading a person actually records must undergo a further step of graduation on top of this: "three forty-seven" assigns the continuous pointer position to a finite set of graduations. That is a second loss. What §1.4 says about "covering every possible reading on the dial" and what §5.2 says about "discrete, graduated" both fall on this second loss. The argument of this section uses only the first loss: even if the graduation were infinitely fine, the dimension reduction itself already rules out globally unique inverse recovery.

The strength of this argument varies in different scenarios. In the case of social indicators, the dimension reduction argument is relatively direct—"academic quality" truly cannot be compressed into a single number, because determining academic quality itself requires the discarded dimensions (context, motivation, subtle differences in originality). In the case of physical timekeeping, the force of the dimension reduction argument is weaker—Coordinated Universal Time (UTC) can indeed be well approximated by a single number, and the "dimension reduction" of a clock is more like a thermometer "discarding" barometric pressure information: technically accurate but not a critique of the thermometer. The core application scenarios of this chapter—AI alignment, financial indicators, academic evaluation—belong to the former category, where the dimension reduction argument has its full force. This chapter discusses evaluation tasks in these settings that require preserving the multidimensional differences described above; if a task requires only a defined low-dimensional quantity, the inability to recover the full state does not by itself establish that the indicator fails.

In *The Computer and the Brain*, von Neumann, from an engineering direction, also identified the fundamental difference between two systems of notation, although his objects of comparison were the symbolic systems of the human brain and the computer, not the true state and the reading system: he pointed out that the system of notation used by the human nervous system is an "essentially statistical" mode of computation, with information carried by the statistical characteristics of pulse trains, fundamentally different in type of notation from the discrete, serialized symbolic system on which digital computers depend (in which the precise position of each token determines the meaning of the information)—a notation in which the system trades lower arithmetical precision for higher logical reliability. This book extends that difference into the engineering-side echo of the dimension-reduction thesis; the extension is this book's—von Neumann's own formulation is the trade-off between precision and reliability, not dimension reduction (von Neumann, 1958, *The Computer and the Brain*,

section “Arithmetical Precision or Logical Reliability, Alternatives”).

§4.4 Summary Table

Table 2. The Three Layers of Irreversibility and Their Solvability

Layer	Predicament	Fiber State	Solvable in Principle?
Layer 1: Information Deficit	The mapping function is unknown	The full set when the mapping is completely unknown and there are no prior constraints; potentially smaller when prior constraints are present	Solvable, requires external anchor
Layer 2: Dynamic Corrosion	Incentives and adaptation may rewrite the mapping function	Existing calibration constraints may fail and require continual updating	Continuous calibration is required; convergence depends on the conditions of corrosion and calibration
Layer 3: Ontological Dimension Reduction	The mapping itself is necessarily lossy	Non-singleton fibers exist that calibration cannot eliminate	Not completely eliminable in principle

The three layers are progressive. Layer 1 says calibration is needed. Layer 2 says that the mapping on which calibration relies may be corroded. Layer 3 says even if calibration is not corroded, the mapping itself is lossy. The progression here is that each later layer points to a limitation that the previous layer’s solution cannot eliminate. In reality, all three layers are usually simultaneously present.[11]

This chapter does not offer a scheme to “break out of the room”—for low-dimensional tasks such as telling the hour, the obstacle lies in layer 1 (the unknown mapping, solvable in principle); but whenever the task requires recovering the dimensions discarded by the reduction, layer 3 is mathematically ineliminable and the room has no complete exit. What this chapter provides is a precise diagnosis of the structure of the predicament, and a route to reduce the risk of catastrophe through “visibility” within that structure.

[11]The Law of Requisite Variety, given by Ashby (1956) in Chapter 11 of *An Introduction to Cybernetics*, provides an external metric statement for layer 1 from cybernetics: the regulator’s capacity to regulate cannot exceed its capacity as a communication channel (original S.11/11); when the outcome is required to remain perfectly constant, and different disturbances still produce different outcomes under a fixed regulatory action, the regulator’s variety must be no less than the disturbance’s variety (S.11/5–S.11/7). If the system itself already blocks some of the disturbance variety, that reduction should be taken into account; the full disturbance variety cannot simply be used as the regulator’s necessary lower bound (S.11/9, S.11/11). What the Clock Room removes is precisely this channel—the clocks are all inside the room, not connected to the outside. The nature of the law requires clarification: Ashby specifically notes that its impossibility has nothing to do with the nature of matter or the nature of machines, but is merely a statement about which arrangements cannot be made in a rectangular table, and therefore owes nothing to experiment (S.11/10). It is a structural proposition, not an empirical one—it will not be violated; under the conditions above, when the regulator’s variety is insufficient to match the disturbance’s variety, the required perfectly constant outcome cannot be achieved—such failure manifests the law’s constraint rather than violating it. The dynamic corrosion of layer 2 exceeds the scope of this law: Ashby treats the disturbance and the table as an exogenously given hard world (S.11/11), and does not address the case where the variety of the disturbance grows endogenously due to the act of regulation; this chapter’s extension of layer 2 onto this metric language is an extension of this chapter, not a proposition of Ashby.

§5 Working on Clocks vs. Working in Time

The three layers of irreversibility describe the relationship between readings and reality. But there is one more issue not within those three layers—the effect of the act of mapping on the cognizer himself.

§5.1 Two Modes of Work

For the vast majority of human civilization's history, human contact with the world was multi-channel, continuous, and relational. The texture felt by the hand touching a material, the eye seeing the color change of a flame, the master demonstrating an action to the apprentice—these are working in time, accumulating the capacity to do in the process of doing.

When reading systems intervene—timekeepers, Key Performance Indicators (KPIs), scores, algorithms—a portion of the work is transferred from time onto the clock. In these settings, some human actions come to rely more on one or a few readings rather than drawing simultaneously on the existing multiplicity of channels. The reading is clearer, more convenient, better suited for coordination. What it brings first is not error but benefit; precisely because the benefit is enormous, its existential cost is easily overlooked. Yet it is also simultaneously reshaping the cognizer's position.

The difference between these two modes of work is not 'primitive vs. modern'—it is not an evolutionary ladder. They are two different modes of making contact with the world, each suitable to its own contexts. Weizenbaum (1976), with his subtitle 'From Judgment to Calculation,' identified the epistemological substance of this distinction: 'Power is nothing if it is not the power to choose. Instrumental reason can make decisions, but there is all the difference between deciding and choosing.'[12] Working in time more directly exemplifies judgment—choosing within embedded, relational, not-fully-formalizable situations; working on clocks more prominently involves calculation—making decisions within a formal system constituted by readings. Both can produce action, and both can be accompanied by the actor's taking responsibility for the meaning of the action; what this chapter criticizes is the situation in which the final reason for action is reduced to the optimal instruction of a reading system and choice is no longer undertaken by the actor. Working in time has long been an important basis of human existence. Over the last two centuries, with industrialization and the expansion of modern organizations, working on clocks has come to occupy an important place in a growing range of coordination and governance settings, contributing to resource exploitation and the Industrial Revolution and giving the predicaments described by the three layers of irreversibility a new scale and institutional form. The question raised by this chapter is not 'whether to use clocks,' but: when 'working on clocks' becomes the overwhelmingly dominant mode, the predicaments of the three layers of irreversibility will systematically intensify.

Articulation with "scientism." Wang Defeng has offered an adjacent diagnosis of "scientism": "To oppose 'scientism' is not to oppose 'science' … Once science breaks beyond its own range of meaning and undertakes the interpretation of the meaning of human life, the result is necessarily the alienation and reification of human

[12] Weizenbaum, Joseph. *Computer Power and Human Reason: From Judgment to Calculation*. San Francisco: W. H. Freeman, 1976. The quoted passage is from Chapter 10, "Against the Imperialism of Instrumental Reason": "Power is nothing if it is not the power to choose. Instrumental reason can make decisions, but there is all the difference between deciding and choosing." See also the discussion of the clock in Chapter 1, "On Tools": "experiences of reality had to be representable as numbers in order to appear legitimate in the eyes of the common wisdom." The subtitle "From Judgment to Calculation" is structurally isomorphic to this chapter's proposition "from the time dimension to the clock dimension."

beings." (author's translation)[13] When "working on clocks" becomes the overwhelmingly dominant mode, its philosophical counterpart is precisely "scientism"—taking measurement as dwelling, taking the reading as the foundation of life. This chapter does not develop this correspondence; it only marks here that the epistemological predicament described by the three layers of irreversibility and the ontological pathology of "scientism" as an existential orientation are, structurally, two faces of the same set of problems.

§5.2 The Time Dimension and the Clock Dimension

This can be stated more precisely:

The time dimension is the continuous, multi-channel, relational mode of existence. The relationship between a person's hunger and their last meal, between their fatigue and their labor, between their heartbeat and their emotions—these are relational, not sensory reception by an isolated individual.

The time dimension is not "pure existence without mapping"—bodily sensations are themselves mappings. Vision passes through multiple layers of processing: the retina, optic nerve, visual cortex; touch passes through receptors, spinal cord, cerebral cortex. Plato had already distinguished "with" from "through": in *Theaetetus* 184c–184d, Socrates asks whether we perceive "with" the eyes and ears or "through" them, and Theaetetus answers "through"; Jowett's translation sums up the distinction as perceiving through, rather than with, the senses. The full antithesis lies in its first half: sensible things are perceived "not through, but with the mind"—the mind occupies the position of "with," the senses that of "through" (Jowett trans.). The causal chain of the body is not necessarily "shorter" than a clock reading. But the body can do something the clock cannot: simultaneous multi-channel integration. A person simultaneously receives signals on independent channels—temperature, light, hunger, fatigue, heart rate, etc.—and these channels can cross-verify each other. The clock is a single-channel reading, with no corroboration. The difference in fiber does not come from the absolute length of the causal chain, but from the number of channels and the possibility of cross-verification.

The advantage of multi-channel cross-verification has a premise: the sources of bias in different channels are usually different. The bias of vision and the bias of touch are usually caused by different factors, so mutual verification is effective. But when the source of bias is the same—a drug altering all perceptual channels, a high fever causing the sense of time, temperature, and fatigue to all be disordered—multi-channel cross-verification fails, because all channels are simultaneously distorted by a single source. The advantage of multi-channel is statistical (usually effective), not structural (necessarily effective).

The clock dimension is the discrete, graduated, reading-only mode of operation. A single number, no corroboration. The relational, multidimensional existence is compressed into a single-dimension numerical reading.

What this chapter argues is not "there exists a mode of being without mapping," but "there are better and worse mappings—on the basis of the existing observations, adding verifiable channels that supply new constraints

[13] Wang Defeng's explicit definition of "scientism" is given in *Searching for Meaning* (Shandong Literature and Art Publishing House, 2022), in the chapter "Contemporary Cultural Conditions and the Life of Chinese Culture," section "Scientism and the Logic of Capital": "To oppose 'scientism' is not to oppose 'science', we must distinguish these two concepts: one is science, the other is scientism. There is nothing wrong with science itself. … But we oppose scientism. Once science breaks beyond its own range of meaning and undertakes the interpretation of the meaning of human life, the result is necessarily the alienation and reification of human beings." The most concise definition appears in the lecture notes of *Introduction to Philosophy* (Wang Defeng, 2014): "Today we believe only what is scientific is real. There is a belief here, and we call this belief scientism. Scientism is a belief. What we criticize is not science but scientism." (All renderings of Wang Defeng's text in this note are the author's translation.) Together these passages give the twofold characterization of "scientism" used in this section: not science itself, which is testable and falsifiable, but the belief that "only the scientific is real"—a belief that elevates the reading from a tool into the foundation of life.

can further shrink the set of compatible states; channels that duplicate information or share the same source of bias do not guarantee this effect." These two modes each suit their own contexts; when multi-channel, cross-verifiable mappings are systematically compressed into single-channel readings, the predicaments of the three layers of irreversibility intensify. This chapter by default compares candidate sets by the partial order of set inclusion: the fiber of A is said to be no larger than that of B only when $F_A \subseteq F_B$, and to shrink only under strict inclusion. Finite sets can be counted, and task-relevant information measures can be computed when a probability model is supplied; this chapter does not presuppose a single numerical scale applicable across all scenarios.

§5.3 Entering the Clock Dimension

When a person begins to rely entirely on the clock-face reading as the basis for decision-making, they have entered the clock dimension from the time dimension. This is not a neutral switch of operational mode—the rhythm of action shifts from endogenous to externally paced (the precise referent of the abstract's slipping from "existing in time" to "reacting within the clock"). This section further discusses the situation in which this mode combines with the incentive and adaptation processes described in §4.2 and experience that has not been encoded no longer enters judgment; in this situation, the three difficulties reinforce one another.

Layer 1: The absence of independent calibration. In the time dimension, multi-channel cross-verification provides a partial constraint on the mapping function—one does not know the mapping in full, but consistency checks across multiple channels can rule out some possible states, and the fiber shrinks in part. The clock dimension is defined as: single-channel, no corroboration, no external access. This means that independent calibration no longer provides new constraints; existing priors and calibration knowledge can still restrict the fiber, but the readings alone cannot confirm that these constraints remain valid. Only when the mapping function is completely unknown and there are no prior constraints is the fiber the full set.

Layer 2: Erosion of calibration by incentives and adaptation. This follows the premises of §4.2: the reading is used for rewards, punishment, resource allocation, or performance evaluation, and actors respond to these arrangements through strategic adjustment or selective adaptation. In this feedback process, the mapping function can no longer be treated as a permanently stable exogenous parameter, and existing calibration must submit to continual re-examination.

Layer 3: Ontological dimension reduction. Compressing a continuous, multidimensional reality into a discrete, finite-dimensional reading is, in information-theoretic terms, necessarily lossy. But ontological dimension reduction is one layer deeper than information loss. Weizenbaum (1976) noted: "experiences of reality had to be representable as numbers in order to appear legitimate in the eyes of the common wisdom." This reveals the legitimacy logic of the reading system: experiences that cannot be quantified are not merely "temporarily unseen"—they lose their standing as legitimate experience. The reading system does not merely add a layer of simplification onto reality—it suppresses what cannot be encoded. Dimensions not encoded by the reading do not exist in the operational language of the clock dimension and are not recognized as legitimate cognitive objects. One cannot ask, in the language of the reading, "what has the reading failed to encode"—this very question lies outside the vocabulary of the clock dimension.

Progression and self-sealing across the three layers. When the discernible information about a deviation disappears along with the omitted dimensions, these deviations cannot be identified from the clock dimension's

readings and their existing constraints alone; signals of readings violating existing constraints may still expose anomalies, but detecting an anomaly does not amount to completing calibration. If one could see the dimensions omitted by the reading, Layer 1 could be partially repaired (one would know what the mapping is missing); if one could see the gaming and adaptive drift unfolding, Layer 2 could be partially resisted (one would know that the mapping is degrading). But when the information needed to identify such deviations is indeed lost through dimension reduction, Layer 3 obscures them; whether they become wholly unidentifiable depends on the particular mapping and the existing constraints. The three layers are not three parallel problems but a single self-sealing trap: Layer 3 protects these deviations from being discovered, while the first two continuously deteriorate one's relation to reality. This is a self-sealing predicament formed under the joint action of the conditions above.

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§6 The Contemporary Clock Room

The three layers of irreversibility are not merely an abstract epistemological predicament. They take different forms in the evaluation, alignment, and use of contemporary AI systems.

§6.1 AI Alignment and the Helpful, Honest, Harmless (HHH) Trilemma

The core framework of contemporary AI ethics—Helpful, Honest, Harmless (HHH)—exposes a deep predicament from the perspective of the Clock Room.

Helpful. When the person in the room faces the question "How long until six o'clock?", he can produce as many precise computational results as there are clocks (N). In computational capacity, he is complete. But he cannot determine which answer corresponds to reality—because the condition for "helpful" is that the answer is aligned with the true time, and he does not know the true time. One of those N answers may happen to be correct, but he cannot identify it among the others. When an AI's training data includes mutually contradictory information sources, if the development and evaluation pipeline implicitly equates Helpful with "providing a definite answer"[14], the system will then forcibly output a "seemingly definite" answer under the condition of an unknown mapping—this is using fluency to mask uncertainty.

Honest. When rewards in the Reinforcement Learning from Human Feedback (RLHF) training of Large Language Models (LLMs) favor confident responses, the system faces pressure to underreport the boundaries of its knowledge. The following studies provide evidence about probability calibration and the imitation of falsehoods, respectively; they do not by themselves establish the same response tendency in all current models. Measurements of the models' capacity for self-assessment show: larger models are well-calibrated on properly formatted multiple-choice and true-or-false questions, but policies trained with RLHF exhibit calibration distortion (this distortion can be partially corrected through temperature adjustment), and models have difficulty calibrating their judgment of "whether they know the answer" on new tasks (Kadavath et al., 2022); models also systematically mimic human misconceptions in their training data (Lin et al., 2022). Paradigms such as Constitutional AI attempt to mitigate this problem through principle-based constraints (Bai et al., 2022), but the

[14] Kalai et al. (2026) analyzed accuracy-based evaluations that score abstention the same as errors: when the model believes there is still a chance of answering correctly, the expected score of guessing exceeds that of abstaining; such metrics form an incentive against abstaining when uncertain, acting through model selection and development.

fundamental difficulty of calibration—the mapping function is unknown—remains independent of any training technique.

Harmless. When an AI gives a fluent, confident, but uncalibrated answer in a high-stakes domain, a surface appearance of harmlessness conceals substantive harm.

Under the narrow operational definitions used here, there exists a structural tension among the three. If Helpful is interpreted as necessarily outputting a definite, actionable answer, then any specific answer has a probability of being wrong; if Honest is interpreted as necessarily declaring uncertainty, then the declaration itself may reduce the helpfulness of the answer; if Harmless is interpreted as avoiding any possible erroneous output, then silence itself violates the requirement of Helpful. Whether the three can be compatible under more refined definitions is a separate question—but two kinds of evidence in current engineering practice point to two respective risks: the calibration distortion noted above suggests that training may distort the expression of confidence; and accuracy-based evaluations that score abstention as incorrect form an incentive favoring definite outputs through model selection and development (Kalai et al., 2026). When a system converts this kind of scoring advantage into an output strategy, the first definition of Helpful may come at the cost of underreporting uncertainty; in high-stakes applications, this trade-off also increases the risk of harm. It should be noted that the proposers of the HHH framework already acknowledged the tension among the three objectives in the original paper (Askell et al., 2021); the subsequent Constitutional AI route attempts to redefine Helpful through principle-based constraints—so that it includes the dimension of honesty, that is, expressing uncertainty when uncertain (Askell et al., 2021; Bai et al., 2022 advances this direction through principle-based constraints). The analysis in this chapter does not negate this direction, but reveals: so long as the system's optimization objective includes a preference reward for "fluent certainty," the system continues to face structural pressure to produce outputs that "appear certain but are in fact uncertain."

§6.2 The Chinese Room and the Clock Room

In his paper "Minds, Brains, and Programs," published in *Behavioral and Brain Sciences*, Searle (1980) designed the Chinese Room thought experiment to test a claim of strong AI: that an appropriately programmed computer can literally understand stories and possess cognitive states, and that the program itself thereby constitutes an explanation of human understanding.

The experiment imagines a person who knows no Chinese being locked in a room. The person follows a set of rules written in English, correlating inputs with outputs solely by the shapes of the symbols, and returns the corresponding Chinese symbols. He does not know that these symbols are respectively stories, questions, and answers, nor does he know that this set of rules is the program. People outside the room cannot distinguish his Chinese answers from those of a native Chinese speaker.

Searle argues on this basis that correct inputs and outputs and purely formal symbol manipulation are insufficient to constitute understanding. This argument targets strong AI, not weak AI, which treats the computer as a tool for investigating the mind.

The core differences between the Chinese Room and the Clock Room are as follows:

Table 3. A Comparison of the Chinese Room and the Clock Room

Dimension	Chinese Room	Clock Room
Rules	Has a manual (guarantees correct output)	Has no manual
Local understanding	Absent (the person does not understand Chinese)	Present (the person understands each clock)
Core proposition	Syntactic operation $\neq$ semantic understanding	Local understanding $+$ all internal readings $\neq$ global truth

The Chinese Room reveals that "correct output" does not entail "understanding." The Clock Room reveals that "understanding" does not entail "correct output." The two are nested, not parallel—regardless of how the dispute over the Chinese Room is resolved, the problem of the Clock Room exists independently. Being able to understand the readings does not mean one already possesses an independent basis for calibrating them.

A common objection to the Chinese Room is the "Systems Reply": even if the person in the room does not understand Chinese, the whole system composed of the person, the rulebook, the symbol database, and the computational tools may nevertheless understand Chinese. Searle's response is to have the person internalize the rules, the database, and the entire computational process; he still does not understand Chinese. Merely enlarging the system's components therefore leaves the original argument unanswered.

The Clock Room is not affected by the "Systems Reply" rebuttal that the Chinese Room encountered—networking all the clocks, adding algorithms and weighted averaging, the entire "system" producing a beautifully polished consensus reading—this remains an instance of the judgment of §1.3: increasing system complexity improves precision, not alignment with reality.

§6.3 The Mediation Problem

Contemporary mediation systems—from recommendation algorithms to large language models—add a new problem: natural-language interfaces may lead some users, in some tasks, to become less aware of the mediating layer.

The person in the Chinese Room knows he is following a rulebook. The person in the Clock Room knows he is reading clocks. But when a mediator presents output in the form of natural-language dialogue, the user may slip from 'using a tool' towards 'trusting a companion'; the interface may thereby obscure the identity of the mediator.

First, exclude a pseudo-problem. Contemporary mediation systems are not clocks in a sealed room—they can connect to sensors and retrieve real-time data. In a sense, they have windows. Humans have always made contact with the world through mediators (books, teachers, maps, instruments); mediation itself is not the problem. The particularity of contemporary AI mediators lies in their interface: presenting output in the form of natural language dialogue and actively filling in the user's cognitive gaps—completing elided background, reconstructing context, generating seemingly coherent reasoning, providing replies with full syntax and tone. This interface makes it difficult for the user to perceive that they are passing through an intermediary layer.

But the problem is not whether it has windows. The problem is: **people may stop actively using their own windows and grow accustomed to seeing through the mediator's window.** What they then acquire is pri-

marily the mediator's report on reality, rather than reality unmediated. The mediator itself is a mapping layer; the rules governing the selection and presentation of key signals are usually set by the system, rather than being inspected or determined item by item by the user.

Forgetting that one is making contact with the world through a mediator is more noteworthy than the bias of the mediator itself.

It must be stated precisely: the problem is not that people cannot check—when a mediation system reports the weather and the user steps outside to find a discrepancy, the user knows the system was wrong. People still have their own sensory experience. The real danger is that people stop checking—this is a behavioral problem (habit formation), not merely a structural problem (system uncheckable). But behavioral problems, when scaled, generate structural consequences: when enough people stop checking, the output of the mediator begins to shape consensus, and consensus in turn reinforces trust in the mediator. Behavioral inertia at the individual level becomes, at the collective level, the self-referencing structure described in §4.2.

Calibration—as per Definition 4 in §3.4—targets more than just the output accuracy of the model. When a mediation system causes a user to lose the habit of independent verification, even if the output is perfectly accurate, the relationship between the user and reality has already changed—they have slipped from working in time to working on clocks.

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§7 Conclusion

§7.1 Self-Reference

This chapter itself is an act of mapping.

If, after reading, the reader feels they have "understood" that the clock is not time—they should ask themselves: is this understanding derived from calibration (they independently verified the argument using their own experience), or from yet another reading (they were persuaded by the narrative of the argument)? The latter is another instance of the closed-loop pseudo-calibration of §3.6.

The entire argument of this chapter can be tested using its own terms—but two different kinds of testing must be distinguished. For the distinction between the "time dimension" and the "clock dimension" in §5, readers can recall their own experiences in concrete situations: when was their judgment wholly reliant on a single numerical reading, and when was it reliant on multi-channel, cross-verifiable bodily perceptions? Whether the distinction holds is for the reader to judge based on their own experience—this does not belong to formal verification. For the mathematical proposition of §3.5 ("the fiber is the full set when the mapping is completely unknown and candidate mappings are subject to no prior constraints"), the method of testing is to examine the logical structure of the proof—this is formal verification. The warning of self-reference still applies to the latter (the reader may merely have been persuaded by the narrative of the proof without actually checking the logic), but the testing path is different.

If, after reading, the reader accepts the conclusion solely because of the argument's narrative force—without tracing the ontological part back to their own experience, and without checking the argument's logic for the formalized part—then the very belief that "I have understood the mapping predicament" is itself a reading: it

comes from the persuasive force of the argument, not from the reader's independent verification. This adds one more internally consistent new clock to the room.

§7.2 Core Judgment

Under the conditions of a complete absence of continuous, external calibration, a set of mapped readings does not by itself guarantee the presence of verifiable information about the object being mapped.

Information can be missing (remediable). Calibration can be corroded (difficult to remedy). The mapping itself can be lossy (irremediable). The act of mapping also changes the cognizer's relationship with the world—slipping from working in time to working on clocks.

By their number alone, hundreds or thousands of clocks cannot answer "what time is it now." Hundreds or thousands of papers cannot answer "what is the truth." Hundreds or thousands of indicators cannot answer "what is the quality."

It is not because they are too few. It is because the multiplicity lies in the mapping space, not in the space of reality.

§7.3 From Epistemology to Bookkeeping

Civilization cannot do without clocks. The question has never been "whether to use clocks," but when people are compelled to use clocks, how to track the gap between the clock and reality? How to sound an alarm before the accumulated gap reaches a dangerous level? How to distinguish between "this clock has a bias that is controllable" and "this clock's bias is being systematically concealed"?

What follows is not a complete solution of this chapter, but a minimal institutional intuition naturally derived from the above analysis. Tracking the gap requires a bookkeeping structure—a set of compulsory disciplines, so that every accepted reading is simultaneously recorded with: how much of the mapping function is known, the extent of the candidate set and any task-specific measure available, when the last calibration occurred, and who bears the consequences of changes in the candidate set.

To address Layer 1, bookkeeping tracks the state of calibration—which mapping functions are known and which are unknown. To address Layer 2, bookkeeping tracks the rate of corrosion—the ratio of the mapping function's drift rate to the calibration frequency, triggering an alarm when the ratio exceeds a threshold. This ratio must be specified within each concrete task—the measure of drift, the effectiveness of calibration, and the alarm threshold; what is proposed here is the tracking relation, not a universal numerical scale presupposed for all scenarios. To address Layer 3, bookkeeping does not track the discrepancy (dimensions not encoded cannot be measured from that reading alone), but marks the existence of the discrepancy—writing beside every record "here there is ineliminable uncertainty—the mapping may have discarded dimensions that are not encoded." The mark does not eliminate this uncertainty, but it prevents people from forgetting it. Forgetting is more dangerous than the uncertainty itself.

The bookkeeping structure is not a solution to the problem. It is a tool for making the problem visible. The difference between "not having bookkeeping" and "having bookkeeping, even though the bookkeeping itself is also a clock" is real—but this difference concerns visibility, not elimination. Bookkeeping lets people know that they are operating in a system with fibers. It does not eliminate the fibers. The bookkeeping structure itself

is also a mapping—when its results enter incentive and resource-allocation processes, it may trigger Layer 2 dynamic corrosion and will also exhibit the dimension reduction and residual uncertainty discussed in Layer 3, and its own fibers are ineliminable, so that it must itself be continuously audited. Establishing it does not mean the problem is solved; it only means the problem is pushed to a level where it can be continuously attended to. To build this bookkeeping structure, to convert epistemological insight into operational discipline, is the task of the next step. The operationalized plan for this bookkeeping structure will be given in the Macro-Cognitive Accounting Framework (MCAF) of Chapter 3.

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